Indian Institute of Technology Madras

DA5400W - Foundations of machine learning

Quiz I

Due date: February 9, 2026, 10 pm

Instruction

1. The exam is a closed-note exam.
2. The student is allowed to use an online calculator available on the portal.

Problems

1. Consider the following data matrix $\mathbf{X}$ with $N = 5$ observations and $p = 3$ variables

$$\mathbf{X} = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 8 \\ 1 & 4 & 9 \\ 3 & 2 & 7 \\ 2 & 3 & 8 \end{bmatrix}$$

The Principal Component Analysis (PCA) is applied to $\mathbf{X}$. Which of the following statements are true?

(A) One principal Component explains 100% variability in the data.

(B) Two principal Components explain 100% variability in the data.

(C) Three principal Components explain 100% variability in the data.

(D) The data lies in one-dimensional subspace spanned by the first Principal Component corresponding to the largest eigenvalue of the covariance matrix

(E) The data lies in a two-dimensional subspace spanned by the two Principal Components corresponding to the two largest eigenvalues of the covariance matrix.

(F) The data lies in a three-dimensional subspace spanned by the three Principal Components corresponding to the three largest eigenvalues of the covariance matrix.

[**Mark: 1**]
Answer: B and E
Type: MSQ

2. In PCA, a data matrix $\mathbf{X}$ is centered by pre-multiplying $\mathbf{X}$ with the centering matrix $\mathbf{H} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}_n\mathbf{1}_n^T$. Consider the following matrix $\mathbf{Y}$, in which the rows are samples and the columns are features:

$$\mathbf{Y} = \begin{bmatrix} 1.25 & -0.75 & 0.50 & 1.20 \\ -0.85 & 1.10 & -1.30 & -0.40 \\ -0.20 & -0.35 & 0.60 & -0.50 \\ -0.20 & 0.00 & 0.20 & -0.30 \end{bmatrix}$$

Select the correct option.

(A) $\mathbf{Y} = \mathbf{HX}$ for some 4×4 data matrix $\mathbf{X}$

(B) $\mathbf{Y} \neq \mathbf{HX}$ for some 4×4 data matrix $\mathbf{X}$

(C) $\mathbf{Y} = 2\mathbf{HX}$ for some 4×4 data matrix $\mathbf{X}$

[**Mark: 1**]
Answer: A
Type: MCQ

3. For a particular sample variance-covariance matrix, the eigenvalues are $1, 2, 3$. The fractional variance explained by the first principal component is ___________. Report your answer as a decimal.

[**Marks 1**]
Answer: 0.5 **to** 0.5
Type: NAT

4. Consider the following 3×3 data matrix $\mathbf{X}$ in which the rows are samples, and the columns are features, and a vector $\mathbf{u}$:

$$\mathbf{X} = \begin{bmatrix} -1.2 & 0.3 & 0.9 \\ 0.4 & -1.1 & 0.7 \\ 0.8 & 0.8 & -1.6 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} 3/5 \\ 4/5 \\ 0 \end{bmatrix}$$

The value of the signed magnitude of the projection of the first sample onto the vector $\mathbf{u}$ is ___________. Report your answer as a decimal.

[**Mark: 1**]
Answer: -0.48 **to** -0.48
Type: NAT

5. A 2-dimensional dataset is transformed using PCA. The covariance matrix $\mathbf{S}$ of the data is given by:

$$\mathbf{S} = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}$$

The percentage of the total variance explained by the first principal component is ___________. Report your answer rounded to two decimal places.

[**Mark: 1**]
Answer: 85.71
Type: NAT

6. Consider a dataset $\mathcal{D}$ with 8 points in a 2D space: $A1(2, 10)$, $A2(2, 5)$, $A3(8, 4)$, $B1(5, 8)$, $B2(7, 5)$, $B3(6, 4)$, $C1(1, 2)$, and $C2(4, 9)$.

The clustering task is to partition these points into $K = 3$ clusters using Euclidean distance. Suppose the initial centroids are chosen as: $m_1 = A1(2, 10)$, $m_2 = B1(5, 8)$, and $m_3 = C1(1, 2)$.

Perform the first iteration of the K-means algorithm. After the points are assigned to their nearest initial centroids and the centroids are recomputed (after one iteration), the y-coordinate of the new centroid m_2 is ___________. Report your answer rounded oof to two decimal places.

[**Marks: 2**]
Answer: 6 to 6
Type: NAT

7. Consider the one-dimensional dataset

$$\{1, 5, 8, 10, 2\}.$$

Apply *agglomerative hierarchical clustering* using Euclidean distance and Complete-link (maximum distance) criterion. Now suppose the dendrogram is *cut at a height (distance threshold) of 5.* After performing all necessary merges and applying the cut at distance 5, the number of clusters =__________.

[**Marks: 2**]
Answer: 2 to 2
Type: NAT

8. Consider the following 2D data points:

$$\begin{aligned} &P1 : (7.0, 6.5), \quad P2 : (5.0, 10.0), \\ &P3 : (6.2, 7.1), \quad P4 : (2.0, 3.1), \\ &P5 : (9.3, 2.4), \quad P6 : (8.5, 1.9), \\ &P7 : (2.8, 3.6). \end{aligned}$$

Apply the DBSCAN algorithm with parameters minPts $= 2$ and $\varepsilon = 1.15$ using Euclidean distance.

After running DBSCAN with the given parameters, the number of clusters formed is __________.

[**Marks: 2**]
Answer: 3 to 3
Type: NAT

9. Consider a random sample X_1, X_2, X_3, X_4 of size $n = 4$ from a Normal distribution $N(\mu, \sigma^2)$. The observed sample values are $\{2, 4, 4, 6\}$. The Maximum Likelihood Estimate of the variance σ^2 __________.

[**Marks 2**]
Answer: 2 to 2
Type: NAT

10. Consider a random sample X_1, X_2, X_3, X_4 from a Uniform distribution on $[0, \theta]$. The observed sample values are $\{3, 5, 7, 9\}$. Using the **Method of Moments**, The estimate of θ is __________.

[**Marks 2**]
Answer: 12
Type: NAT